AMD Instinct MI210

Contents

Print to PDF

The AMD Instinct™ MI210 GPU is a mainstream HPC and AI PCIe-form-factor accelerator. This document provides MI210-specific prerequisites, health checks, validation steps, and performance acceptance criteria.

Overview

The AMD Instinct MI210 brings second-generation CDNA architecture to a standard full-height, full-length, dual-slot PCIe® add-in card aimed at single-server HPC and AI deployments. Each MI210 provides 104 compute units, 64 GB of HBM2e memory at up to 1.6 TB/s of memory bandwidth, and up to three AMD Infinity Fabric™ links that enable direct GPU-to-GPU connectivity in dual- and quad-GPU hive configurations. The card is passively cooled with a 300 W TDP and supports PCIe® Gen4 host connectivity.

The MI210 is built on AMD CDNA 2 architecture (gfx90a) in a PCIe add-in-card form factor with 104 compute units and 64 GB of HBM2e memory per accelerator. Unlike the OAM-based AMD Instinct MI250 and MI250X, MI210 deployments are PCIe-attached; GPU-to-GPU traffic uses AMD Infinity Fabric™ links when an Infinity Fabric bridge is installed, otherwise it traverses host PCIe.

- **MI210 Product Page**

- **MI200 Series Data Sheet**
- **MI200 Series Microarchitecture**

System requirements

Operating system support

For the most up-to-date information on supported operating systems and distributions, see the official ROCm documentation:

ROCm System Requirements - Supported Distributions

> **Note**
>
> ROCm docs is the single source of truth for supported versions, distribution compatibility, and required dependencies for the ROCm toolkit.

For general NUMA and OS-level tuning that applies to all AMD Instinct hosts, see OS tuning. MI210 systems share the CDNA 2 BIOS baseline with MI250 (EPYC 7003-series); MI210-specific values are in the BIOS settings section below. Power limits and xGMI topology entries may differ from MI250 — consult your platform vendor's BIOS guide for MI210-specific values.

GPU identification

All MI210 GPUs (PCI vendor:device `1002:740f`) should appear in `lspci` output:

```
sudo lspci -d 1002:740f
```

Expected output example:

```
03:00.0 Display controller: Advanced Micro Devices, Inc. [AMD/ATI] Aldebaran/MI
27:00.0 Display controller: Advanced Micro Devices, Inc. [AMD/ATI] Aldebaran/MI
43:00.0 Display controller: Advanced Micro Devices, Inc. [AMD/ATI] Aldebaran/MI
63:00.0 Display controller: Advanced Micro Devices, Inc. [AMD/ATI] Aldebaran/MI
83:00.0 Display controller: Advanced Micro Devices, Inc. [AMD/ATI] Aldebaran/MI
a3:00.0 Display controller: Advanced Micro Devices, Inc. [AMD/ATI] Aldebaran/MI
c3:00.0 Display controller: Advanced Micro Devices, Inc. [AMD/ATI] Aldebaran/MI
e3:00.0 Display controller: Advanced Micro Devices, Inc. [AMD/ATI] Aldebaran/MI
```

BIOS settings

For maximum MI210 GPU performance on systems with AMD EPYC™ 7003-series processors (codename “Milan”) and AMI System BIOS, the following BIOS settings have been validated. These settings should be set as default values in the system BIOS. Analogous settings for other non-AMI BIOS providers can be configured similarly. On systems with Intel processors, some settings may not apply.

> **Note**
>
> The BIOS setting locations and names may vary by hardware and BIOS vendor. Consult your system documentation for details.

BIOS setting location	Parameter	Value	Comments
Advanced / PCI subsystem settings	Above 4G decoding	Enabled	GPU large BAR support.
AMD CBS / CPU common options	Global C-state control	Auto	Global C-states.
AMD CBS / CPU common options	CCD/Core/Thread enablement	Accept	May be necessary to enable the SMT control menu.
AMD CBS / CPU common options / performance	SMT control	Disable	Set to Auto if the primary application is not compute-bound.
AMD CBS / DF common options / memory addressing	NUMA nodes per socket	NPS 1, 2, or 4	See NPS memory configuration below.
AMD CBS / DF common options / memory addressing	Memory interleaving	Auto	Depends on NPS setting.
AMD CBS / DF common options / link	4-link xGMI max speed	18 Gbps	Set to highest rate supported by the CPU.
AMD CBS / NBIO common options	IOMMU	Disabled	
AMD CBS / NBIO common options	PCIe ten bit tag support	Auto	
AMD CBS / NBIO common options	Preferred IO	Bus	
AMD CBS / NBIO common options	Preferred IO bus	Use `lspci` to find PCIe device ID	

BIOS setting location	Parameter	Value	Comments
AMD CBS / NBIO common options	Enhanced Preferred IO mode	Enabled	
AMD CBS / NBIO common options / SMU common options	Determinism control	Manual	
AMD CBS / NBIO common options / SMU common options	Determinism slider	Power	
AMD CBS / NBIO common options / SMU common options	cTDP control	Manual	Set cTDP to the maximum supported by the installed CPU.
AMD CBS / NBIO common options / SMU common options	cTDP	Varies by CPU	Value in watts. Consult your platform vendor.
AMD CBS / NBIO common options / SMU common options	Package power limit control	Manual	Set to the maximum supported by the installed CPU.
AMD CBS / NBIO common options / SMU common options	Package power limit	Varies by CPU	Value in watts. Consult your platform vendor.
AMD CBS / NBIO common options / SMU common options	xGMI link width control	Manual	
AMD CBS / NBIO common options / SMU common options	xGMI force link width	2	0: x2 / 1: x8 / 2: x16
AMD CBS / NBIO common options / SMU common options	xGMI force link width control	Force	

BIOS setting location	Parameter	Value	Comments
AMD CBS / NBIO common options / SMU common options	APBDIS	1	Disable DF P-states.
AMD CBS / NBIO common options / SMU common options	DF C-states	Enabled	
AMD CBS / NBIO common options / SMU common options	Fixed SOC P-state	P0	
AMD CBS / UMC common options / DDR4 common options	Enforce POR	Accept	
AMD CBS / UMC common options / DDR4 common options / Enforce POR	Overclock	Enabled	
AMD CBS / UMC common options / DDR4 common options / Enforce POR	Memory clock speed	1600 MHz	Set to max speed if using 3200 MHz DIMMs.
AMD CBS / UMC common options / DDR4 common options / DRAM controller configuration / DRAM power options	Power down enable	Disabled	RAM power down.
AMD CBS / security	TSME	Disabled	Memory encryption.

NBIO link clock frequency

The NBIOs (4× per AMD EPYC™ processor) are the serializers/deserializers (SerDes) that convert and prepare I/O signals for the processor’s 128 external I/O interface lanes (32 per

NBIO). The NBIO link clock frequency (LCLK) controls the speed of the internal bus connecting NBIO silicon to the data fabric. All data between the processor and PCIe lanes flows through the data fabric at this frequency, so it must be forced to the maximum for optimal PCIe performance.

For AMD EPYC™ 7003-series processors, configuring all NBIOs to **Enhanced Preferred I/O** mode (see BIOS table above) is sufficient to enable the highest LCLK frequency. The AMD-IOPM-UTIL utility required for EPYC 7002 (MI100) is not needed on EPYC 7003 systems.

NPS memory configuration

For the number of NUMA nodes per socket (NPS), follow the guidance of the HPC Tuning Guide for AMD EPYC™ 7003 Series Processors for the optimal host configuration. For most HPC workloads, NPS4 is the recommended value.

Acceptance criteria

The MI210 system acceptance process validates that the platform is correctly configured, stable, and performing to expectations. Follow the sequence: Prerequisites → Basic Health Checks → System Validation → Performance Benchmarks.

System acceptance process

1. **Prerequisites validation** - Ensure all system requirements and dependencies are met
2. **Basic health checks** - Verify hardware detection and basic system health
3. **System validation** - Conduct comprehensive stress testing and qualification
4. **Performance benchmarks** - Validate compute, memory, and interconnect performance

The system is accepted when all criteria below are successfully validated.

Prerequisites validation

Ensure all system requirements are met before proceeding with validation. See the Prerequisites documentation and System setup for more details.

- ✅ Supported operating system version installed
- ✅ Compatible ROCm version installed
- ✅ BIOS configured per MI210 BIOS settings (EPYC 7003-series baseline; verify power limits with platform vendor)
- ✅ Required kernel parameters present: `pci=realloc=off`, `pci=bfsort`, `iommu=pt`, and `amd_iommu=on` (or `intel_iommu=on` on Intel hosts) — see Kernel Parameters
- ✅ Minimum 512G system memory available
- ✅ Latest applicable firmware applied consistently across nodes
- ✅ ROCm Validation Suite (RVS) installed

Basic health checks

These checks ensure fundamental system health and proper GPU detection. For detailed procedures, see Health Checks.

Test	Command	Pass/Fail criteria
Check OS distribution	`cat /etc/os-release`	**Pass**: OS version listed in compatibility matrix **Fail**: Otherwise
Check kernel boot arguments	`cat /proc/cmdline`	**Pass**: Contains `pci=realloc=off`, `pci=bfsort`, `iommu=pt`, and `amd_iommu=on` or `intel_iommu=on` **Fail**: Otherwise
Check for driver errors	`sudo dmesg -T | grep amdgpu | grep -i error`	**Pass**: Null **Fail**: Errors reported
Check available memory	`lsmem | grep "Total online memory"`	**Pass**: ≥ 512G **Fail**: Less than 512G
Check GPU presence	`sudo lspci -d 1002:740f`	**Pass**: 4 MI210 GPUs found **Fail**: Otherwise
Check GPU link speed and width	`sudo lspci -d 1002:740f -vvv | grep -e DevSta -e LnkSta`	**Pass**: Speed 16GT/s, width `x16`, no `FatalErr+` **Fail**: Otherwise
Monitor utilization metrics	`amd-smi monitor -putm`	**Pass**: Idle metrics as specified **Fail**: Otherwise
Check system kernel logs for errors	`sudo dmesg -T | grep -i 'error|warn|fail|exception'`	**Pass**: Null **Fail**: Otherwise

System validation

Comprehensive validation ensures system stability under load. For detailed procedures, see System Validation.

Test	Command	Pass/Fail criteria
Compute/GPU properties	`rvs -c ${RVS_CONF}/gpup_single.conf`	**Pass**: All GPUs listed with no errors **Fail**: Missing GPUs or errors
GPU stress test (GST)	`rvs -c ${RVS_CONF}/MI210/gst_single.conf`	**Pass**: `met: TRUE` in logs **Fail**: Target GFLOP/s not met
Input energy delay product (IET)	`rvs -c ${RVS_CONF}/MI210/iet_single.conf`	**Pass**: `met: TRUE` for all actions **Fail**: Otherwise
Memory test (MEM)	`rvs -c ${RVS_CONF}/mem.conf -l mem.txt`	**Pass**: All tests passed; bandwidth ~1.1TB/s per GPU **Fail**: Any test failed or low bandwidth
PCIe bandwidth benchmark (PEBB)	`rvs -c ${RVS_CONF}/MI210/pebb_single.conf`	**Pass**: All distances and bandwidths displayed **Fail**: Missing data
PCIe qualification tool (PEQT)	`rvs -c ${RVS_CONF}/peqt_single.conf`	**Pass**: All actions true **Fail**: Otherwise
P2P benchmark and qualification tool (PBQT)	`rvs -c ${RVS_CONF}/pbqt_single.conf`	**Pass**: `peers:true` lines and non-zero throughput **Fail**: Otherwise

Performance benchmarks

Performance validation ensures the system meets MI210 specifications. For detailed procedures, see Performance Benchmarking.

TransferBench all-to-all

Command: `TransferBench a2a`

Pass: ≥ 80 GB/s per GPU aggregate

Fail: otherwise

TransferBench peer-to-peer

Command: `TransferBench p2p`

Test	Pass Criteria
UniDir	≥ 35 GB/s per same-socket peer-pair
BiDir	≥ 65 GB/s per same-socket peer-pair (combined)

Fail: otherwise

RCCL Allreduce

Command: `build/all_reduce_perf -b 8 -e 8G -f 2 -g <N>`

Config	Pass Criteria
`-g 4` (single-socket quad)	≥ 30 GB/s avg bus bandwidth
`-g 8` (dual-socket, cross-socket ring)	≥ 8 GB/s avg bus bandwidth

Fail: otherwise

rocBLAS FP32

Command: `rocblas-bench` **(see code block below)**

```
rocblas-bench -f gemm \
  -r s -m 4000 -n 4000 -k 4000 \
  --lda 4000 --ldb 4000 --ldc 4000 \
  --transposeA N --transposeB T
```

Pass: ≥ 28000 GFLOPS

Fail: otherwise

BabelStream

Command: `mpiexec -n 4 wrapper.sh`

Kernel	Threshold (MB/s)
Copy	≥ 1,230,000
Mul	≥ 1,225,000
Add	≥ 1,115,000
Triad	≥ 1,115,000
Dot	≥ 1,170,000

Fail: otherwise